Aerodynamic & Performance Analysis — Single-Engine Gasoline UAV

Prepared: 2026-06-14 · **Configuration:** high-aspect rectangular wing, twin-boom-style tubular fuselage, single tractor (puller) gasoline engine (DLE-120), 28×10 propeller.

> **Read this first.** This is a first-order analytical estimate built from the supplied data sheet using standard handbook methods (component drag build-up, lifting-line induced drag, ISA atmosphere, energy-based fuel model). It is intended for design-trade and sanity-check purposes — **not** a substitute for wind-tunnel data, CFD, or flight test. All assumptions are stated explicitly so any number can be re-derived or corrected. Where the supplied data sheet is internally inconsistent, this is flagged.

Executive summary

Question	Answer (first-order)
Baseline aerodynamics (4 m span, 2 m² wing, AR 8)	$C_D = 0.029 + 0.050 \cdot C_L^2$; **(L/D)max ≈ 13.1**; stall 65 km/h @500 m
Spec consistency	Self-consistent: 600 km ferry = cruising V_{nom} 150 km/h; 10 h endurance at low speed; V_{min} 81 km/h ≈ $1.3 \cdot V_{stall}$; V_{ne} is a structural limit
Max MTOW	50 kg is structure/wing-loading-limited, not power-limited (T/W ≈ 0.52); overload to ~60 kg flyable at reduced g-margin
Best airfoil	**NACA 2415** (15 % thick = cheap deep spar, low pitching moment, low cruise drag); XFOIL-validated. Add flaps for low speed, not more camber
Half-size wing, 4 m span (AR 16)	Keeps (L/D)max ≈ 13; ~18 % more fast-cruise range; stall rises to 87 km/h
Short wing 2 m × 0.5 m (AR 4) — current	Low AR ⇒ **(L/D)max ≈ 7.3**; max speed **power-limited ~205 km/h** so **V_{ne} 240 km/h is unreachable**; cheapest/strongest structure
Current stall speed (1 m² wing, 50 kg)	**≈ 88 km/h @500 m, ≈ 105 km/h @4000 m** (clean); ~75 km/h with flaps
500 km mission (AR 4 wing)	130 km/h econ (12 L) … 178 km/h flat-out (15 L)
Propeller for 160 km/h	28×10 can't do it (pitch-limited ~107 km/h). **Optimum ≈ 28×24 @ ~5000 rpm**, η_p ≈ 0.88, ~3.4 L/h, **~700 km range**

Contents

1. Input parameters · 2. Method & assumptions · 3. Drag polar & key figures · 4. Requested tables (drag / fuel / AoA vs speed) · 5. Max MTOW · 6. Performance summary · 7. Caveats · 8. Cost-effective structure · 9. NACA airfoil optimization · 10. NACA 2415 vs 4415 head-to-head · 11. XFOIL-grade validation · 12. Half-size wing scenario · 13. Recalculation: 2 m span (AR 4), V_{ne} 240 · 14. Propeller pitch & RPM optimization

1. Input parameters (as supplied)

Parameter	Value	Parameter	Value
Wingspan b	4.0 m	MTOW	50 kg
Wing area S	2.0 m² (4.0 × 0.5 m, rectangular)	Empty weight	19 kg
Root = tip chord	0.50 m	Max payload	20 kg
Airfoil	"NACA 25112" (see §2.3)	Fuel capacity	15 L
Engine	DLE-120 (120 cm³ gasoline twin)	Cruise burn	60 mL/min = 3.6 L/h
Propeller	28×10 tractor	V_min	81 km/h (22.5 m/s)
Static thrust	26 kgf (≈255 N)	V_nom	150 km/h (41.7 m/s)
Fuselage pod	Ø160 mm × 1.8 m	V_ne	180 km/h (50.0 m/s)
Tail boom	Ø100 mm × 1.0 m	Load factor	+4.8 / −2.4 g
Max endurance	10 h	Ferry range	600 km
Service altitude	500 m / 4000 m AMSL	Max altitude	5000 m

1.1 Derived geometry & loading

Quantity	Value	Note
Aspect ratio AR = b²/S	**8.0**	moderate-high; good for efficiency
Mean chord	0.50 m	constant (rectangular)
Wing loading W/S @ MTOW	**245 N/m² = 25.0 kg/m²**	high for this class → fast, but high stall speed
Power loading @ ~12 hp	**4.17 kg/hp**	well powered
Static thrust-to-weight	**0.52**	very strong for a fixed-wing
Fuel mass (gasoline ρ≈0.72)	10.8 kg	19 + 10.8 + 20 ≈ 49.8 kg ✔ closes to MTOW

The mass budget closes almost exactly: **19 kg empty + ~10.8 kg fuel + 20 kg payload ≈ 50 kg MTOW.** The airframe is therefore designed to 50 kg, not arbitrarily limited there.

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2. Method & assumptions

2.1 Atmosphere (ISA)

Altitude	Temp	Density ρ	μ (Pa·s)	Speed of sound
0 m (ref)	15.0 °C	1.2250 kg/m³	1.789e-5	340.3 m/s
500 m	11.8 °C	**1.1673 kg/m³**	1.774e-5	338.4 m/s
4000 m	−11.0 °C	**0.8191 kg/m³**	1.661e-5	324.6 m/s
5000 m	−17.5 °C	0.7361 kg/m³	1.628e-5	320.5 m/s

Density ratio σ(4000 m)/σ(SL) = 0.669 — this is the single biggest driver of the differences between the two altitude cases.

2.2 Drag model

Parasite drag from a **component build-up** (Hoerner / Raymer form factors, turbulent flat-plate skin friction `Cf = 0.455/(log₁₀Re)^2.58`). Induced drag from `C_Di = k·C_L²` with `k = 1/(π·e·AR) = 0.0497`, Oswald efficiency **e = 0.80** (rectangular wing, Raymer estimate 0.81, derated for fuselage interference & trim).

Drag polar: **$C_D = 0.0293 + 0.0497 \cdot C_L^2$**

Parasite-drag build-up (at 500 m, 41.7 m/s — C_{D0} referenced to $S = 2$ m²)

Component	Re	C_f	Form factor	Q	S_wet (m²)	ΔC_D0
Wing (exposed)	1.37e6	0.00422	1.345	1.00	3.92	0.0111
Fuselage pod Ø160×1800	4.94e6	0.00337	1.070	1.00	0.95	0.0017
Tail boom Ø100×1000	2.74e6	0.00373	1.085	1.00	0.31	0.0006
Horizontal tail (~0.30 m²)	0.82e6	0.00464	1.280	1.05	0.62	0.0019
Vertical tail (~0.20 m²)	0.82e6	0.00464	1.280	1.05	0.41	0.0013
Engine cooling + exposed cylinders	—	—	—	—	—	0.0060
Fixed landing gear	—	—	—	—	—	0.0035
Misc / excrescence / leakage (+12%)	—	—	—	—	—	0.0031
Total C_D0						**0.0293**

The exposed air-cooled engine, fixed gear and large prop dominate "non-wing" drag — typical for this hobby-gas-engine class. A cowled/cleaned installation could plausibly reach $C_{D0} \approx 0.022$ (see §8).

2.3 Lift model & the airfoil

- 3-D lift-curve slope: `CLα = a₀/(1+a₀/(π·e·AR)) = 4.787/rad = 0.0836/deg`.
- Zero-lift angle $\alpha_0 \approx -2°$ (cambered section); **$C_L = 0.0836 \cdot (\alpha + 2°)$**.
- Assumed **$C_{Lmax} = 1.40$** (clean, no flaps; consistent with a ~12 %-thick cambered section at the wing Reynolds numbers here, ~$0.8–1.4 \times 10^6$).

> **Airfoil note:** "NACA 25112" is not a standard NACA designation. Read as a 5-digit family it parses to design $C_L \approx 0.3$, max camber far aft (~25 % chord) and **12 % thickness** — most likely a transcription of a NACA 5-digit section (e.g. 23012) or "2412". A 12 %-thick cambered section is assumed throughout. If the true section differs, C_{Lmax}, α_0 and C_{D0}(wing) shift, but the trends below hold.

2.4 Propulsion & fuel model

- Shaft power: DLE-120 ≈ **12 hp (8.8 kW)** at sea level; normally-aspirated lapse with density `P ∝ σ`.
- Propeller efficiency $\eta_p \approx 0.70$ in cruise (static thrust 255 N reproduces the 26 kgf figure).
- **Fuel model:** fuel flow ∝ thrust power, calibrated to the manufacturer's quoted cruise point (3.6 L/h at V_nom). This yields an effective overall efficiency **$\eta_{overall} \approx 0.087$** (fuel-energy → thrust-power), typical for a part-throttle 2-stroke gasoline engine. Gasoline energy density 31.25 MJ/L. Fuel flow then scales with required power at every speed.

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3. Drag polar & key aerodynamic figures

Figure	Value	At
Zero-lift drag C_D0	0.0293	—
Induced factor k	0.0497	—
(L/D)_max	**13.1**	C_L = 0.77
Speed for (L/D)max @ 500 m	23.4 m/s (84 km/h)	best range
C_L for min power (max endurance)	1.33	near stall → use V_min
Stall speed @ 500 m (MTOW)	62 km/h (17.3 m/s)	C_Lmax=1.4
Stall speed @ 4000 m (MTOW)	74 km/h (20.7 m/s)	C_Lmax=1.4
Maneuvering speed V_A @ 500 m (+4.8 g)	137 km/h	corner speed

3.1 Consistency check against the supplied data sheet

Spec claim	Model says	Verdict
Ferry range 600 km	At V_nom (150 km/h) fuel = 0.0248 L/km → 15 L ⇒ 605 km, no reserve	✔ matches almost exactly
Max endurance 10 h	Min fuel flow ≈ 1.12 L/h ⇒ 13.4 h still-air	✔ achievable at low speed
V_min 81 km/h	Aerodynamic stall is 62 km/h; 81 km/h = 1.30·V_stall	✔ V_min is a safe operating min, not the stall
V_ne 180 km/h	Power allows ~198 km/h at 500 m → **V_ne is a structural limit**, not power	✔ consistent

The data sheet is therefore self-consistent once you recognise that **V_min (81 km/h) is an operating minimum with ~30 % stall margin**, and the quoted cruise burn corresponds to flying at V_nom.

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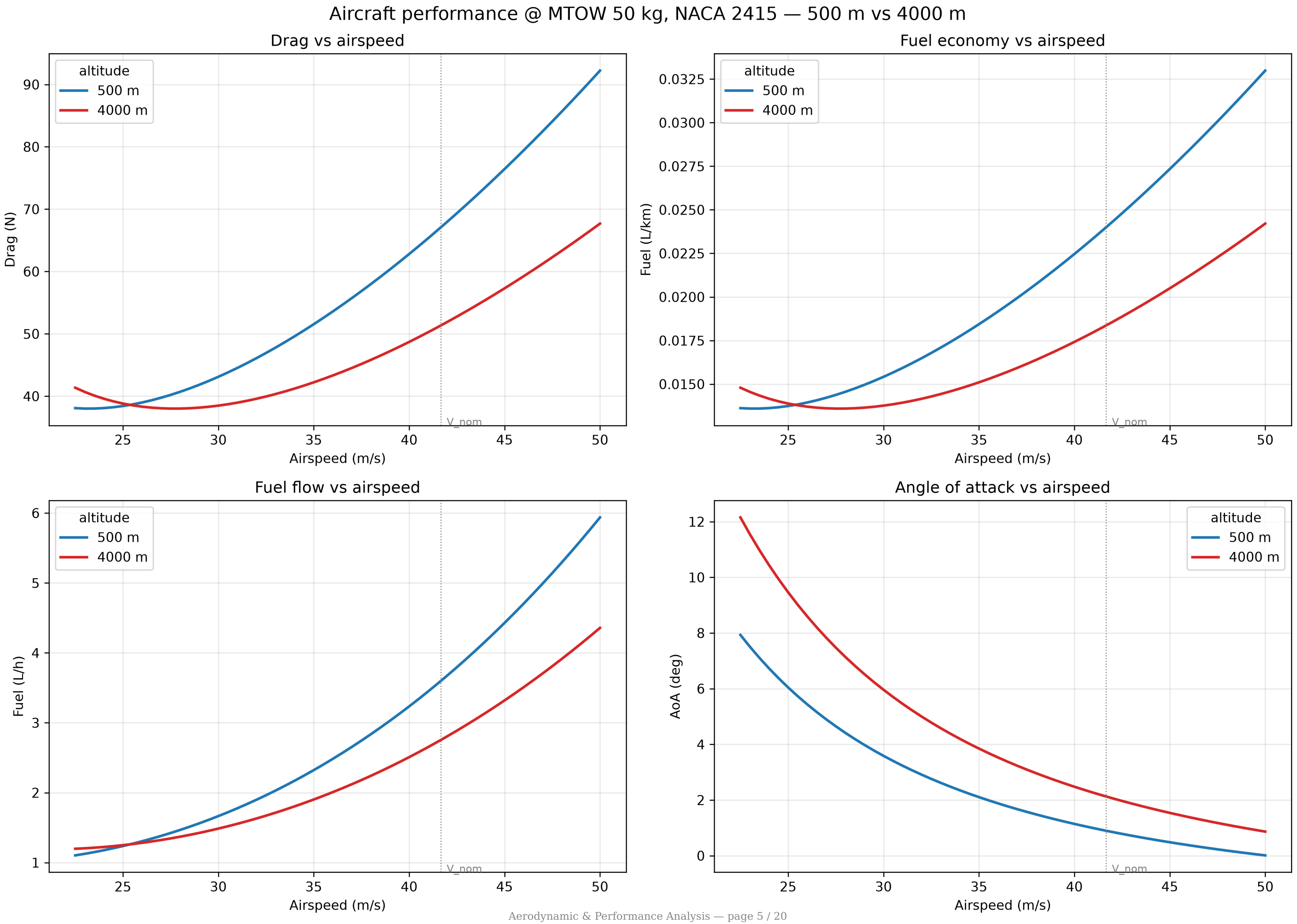
4. Requested tables

Reference condition: **MTOW = 50 kg**, steady level flight (L = W), speed range V_min → V_ne. Each table is given for **500 m** and **4000 m**. AoA is geometric wing angle of attack ($\alpha_0 = -2°$).


Aircraft performance @ MTOW 50 kg, NACA 2415 — 500 m vs 4000 m
Drag vs airspeed
Drag (N)
Airspeed (m/s)
Fuel economy vs airspeed
Fuel (L/km)
Airspeed (m/s)
Fuel flow vs airspeed
Fuel (L/h)
Airspeed (m/s)
Angle of attack vs airspeed
AoA (deg)
Airspeed (m/s)
altitude
500 m
4000 m
V_nom

4.1 Drag vs airspeed

500 m

V (m/s)	V (km/h)	q (Pa)	C_L	C_D	L/D	Drag (N)	AoA (deg)	P_req (kW)	Fuel (L/h)	Fuel (L/km)
22.5	81	295	0.830	0.0635	13.06	37.5	7.93	0.845	1.12	0.0138
25.0	90	365	0.672	0.0518	12.98	37.8	6.04	0.944	1.25	0.0139
27.5	99	441	0.555	0.0446	12.44	39.4	4.65	1.084	1.43	0.0145
30.0	108	525	0.467	0.0401	11.63	42.2	3.59	1.265	1.67	0.0155
32.5	117	616	0.398	0.0372	10.70	45.8	2.76	1.489	1.97	0.0168
35.0	126	715	0.343	0.0351	9.76	50.3	2.10	1.759	2.33	0.0185
37.5	135	821	0.299	0.0337	8.85	55.4	1.58	2.077	2.75	0.0204
40.0	144	934	0.263	0.0327	8.02	61.1	1.14	2.445	3.24	0.0225
42.5	153	1054	0.233	0.0320	7.27	67.4	0.78	2.867	3.79	0.0248
45.0	162	1182	0.207	0.0314	6.60	74.3	0.48	3.344	4.43	0.0273
47.5	171	1317	0.186	0.0310	6.00	81.7	0.23	3.881	5.14	0.0300
50.0	180	1459	0.168	0.0307	5.47	89.6	0.01	4.480	5.93	0.0329

4000 m

V (m/s)	V (km/h)	q (Pa)	C_L	C_D	L/D	Drag (N)	AoA (deg)	P_req (kW)	Fuel (L/h)	Fuel (L/km)
22.5	81	207	1.182	0.0988	11.96	41.0	12.15	0.922	1.22	0.0151
25.0	90	256	0.958	0.0749	12.78	38.4	9.46	0.959	1.27	0.0141
27.5	99	310	0.792	0.0605	13.09	37.5	7.47	1.030	1.36	0.0138
30.0	108	369	0.665	0.0513	12.96	37.8	5.96	1.135	1.50	0.0139
32.5	117	433	0.567	0.0453	12.52	39.2	4.78	1.273	1.68	0.0144
35.0	126	502	0.489	0.0412	11.87	41.3	3.85	1.446	1.91	0.0152
37.5	135	576	0.426	0.0383	11.11	44.1	3.09	1.655	2.19	0.0162
40.0	144	655	0.374	0.0363	10.32	47.5	2.48	1.901	2.52	0.0175
42.5	153	740	0.331	0.0348	9.53	51.4	1.97	2.186	2.89	0.0189
45.0	162	829	0.296	0.0336	8.79	55.8	1.54	2.511	3.32	0.0205
47.5	171	924	0.265	0.0328	8.09	60.6	1.18	2.880	3.81	0.0223
50.0	180	1024	0.239	0.0322	7.45	65.8	0.87	3.292	4.36	0.0242

> The four requested quantities (Drag, Fuel L/km, Fuel L/h, AoA) are all columns of the same two tables above, so they are presented once together rather than repeated. The sections below pull each out explicitly for clarity.

4.2 Drag (N) vs airspeed — extracted

V (m/s)	Drag @500 m (N)	Drag @4000 m (N)
22.5	37.5	41.0
25.0	37.8	38.4
27.5	39.4	37.5
30.0	42.2	37.8
32.5	45.8	39.2
35.0	50.3	41.3
37.5	55.4	44.1
40.0	61.1	47.5
42.5	67.4	51.4
45.0	74.3	55.8
47.5	81.7	60.6
50.0	89.6	65.8

Drag is minimum near (L/D)max. At 500 m the drag bucket sits below V_min, so within the operating band drag rises monotonically with speed. At 4000 m the lower density pushes the bucket up into the band — minimum drag occurs around 27–30 m/s.

4.3 Efficiency — Fuel (L/km) vs airspeed

V (m/s)	V (km/h)	L/km @500 m	L/km @4000 m	km per L @500 m
22.5	81	0.0138	0.0151	72
25.0	90	0.0139	0.0141	72
27.5	99	0.0145	0.0138	69
30.0	108	0.0155	0.0139	65
32.5	117	0.0168	0.0144	59
35.0	126	0.0185	0.0152	54
37.5	135	0.0204	0.0162	49
40.0	144	0.0225	0.0175	44
42.5	153	0.0248	0.0189	40
45.0	162	0.0273	0.0205	37
47.5	171	0.0300	0.0223	33
50.0	180	0.0329	0.0242	30

Best (lowest L/km = best range) is at the (L/D)max speed. At 4000 m the most efficient speed is higher (≈27–30 m/s) but the best achievable L/km is essentially the same as at 500 m — fuel-per-distance depends on L/D, which is altitude-independent, only the speed at which you achieve it changes.

4.4 Power required — Fuel (L/h) vs airspeed

V (m/s)	V (km/h)	P_req @500 m (kW)	Fuel @500 m (L/h)	P_req @4000 m (kW)	Fuel @4000 m (L/h)
22.5	81	0.845	1.12	0.922	1.22
25.0	90	0.944	1.25	0.959	1.27
27.5	99	1.084	1.43	1.030	1.36
30.0	108	1.265	1.67	1.135	1.50
32.5	117	1.489	1.97	1.273	1.68
35.0	126	1.759	2.33	1.446	1.91
37.5	135	2.077	2.75	1.655	2.19
40.0	144	2.445	3.24	1.901	2.52
42.5	153	2.867	3.79	2.186	2.89
45.0	162	3.344	4.43	2.511	3.32
47.5	171	3.881	5.14	2.880	3.81
50.0	180	4.480	5.93	3.292	4.36

Fuel-per-hour tracks power required ($\propto V^3$ at the high-speed end). Loiter/endurance is cheapest at the low-speed end (≈1.1 L/h); high-speed dash at V_ne costs ≈5.9 L/h. At 4000 m fuel flow at a given TAS is lower than at 500 m across most of the band (lower parasite power from lower density), but the engine also makes less power there.

4.5 Angle of attack (deg) vs airspeed

V (m/s)	V (km/h)	AoA @500 m (deg)	AoA @4000 m (deg)
22.5	81	7.93	12.15
25.0	90	6.04	9.46
27.5	99	4.65	7.47
30.0	108	3.59	5.96
32.5	117	2.76	4.78
35.0	126	2.10	3.85
37.5	135	1.58	3.09
40.0	144	1.14	2.48
42.5	153	0.78	1.97
45.0	162	0.48	1.54
47.5	171	0.23	1.18
50.0	180	0.01	0.87

To hold altitude at 4000 m the wing must fly at higher C_L (lower density) ⇒ AoA is several degrees higher than at 500 m for the same TAS. At V_min and 4000 m the AoA (12.2°) is approaching the stall AoA (~13–14°), so V_min should be treated as a hard floor at altitude.

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5. Maximum take-off weight (Max MTOW)

The supplied MTOW is 50 kg. The question is what actually limits it. Computed climb performance vs gross weight:

Gross weight	Best RoC @500 m	Best RoC @4000 m	Stall speed @500 m	Available load factor*
40 kg	13.5 m/s	8.8 m/s	56 km/h	+6.0 g
50 kg (MTOW)	**10.4 m/s**	**6.5 m/s**	**62 km/h**	**+4.8 g**
60 kg	8.3 m/s	5.0 m/s	68 km/h	+4.0 g
70 kg	6.7 m/s	3.8 m/s	74 km/h	+3.4 g

*Available load factor = (structure sized for 50 kg × 4.8 g = 240 kgf limit load) ÷ gross weight.

Findings

1. **Propulsion is not the limit.** With T/W ≈ 0.52 static and ~12 hp on 50 kg, climb is strong (>10 m/s at 500 m) and the aircraft has positive climb beyond 9000 m — the **5000 m "max altitude" is an operational/regulatory ceiling, not a performance ceiling.** Even at 70 kg the climb is healthy.
2. **The binding limits are structural and aerodynamic, not power:** - Structure: the airframe is sized to 240 kgf limit load (50 kg × +4.8 g). Flying heavier erodes the g-envelope — at 60 kg the same structure gives only +4.0 g, at 70 kg +3.4 g. - Stall / V_min: higher weight raises stall speed (and hence take-off/landing speeds and field length). Keeping the stated V_min = 81 km/h as a 1.3·V_stall margin caps weight near 50 kg. - Take-off roll @500 m: ~49 m at 50 kg, ~72 m at 60 kg, ~100 m at 70 kg — all modest.

Conclusion — recommended Max MTOW: - **50 kg is the correct design MTOW** — it is set by the structural +4.8/−2.4 g envelope and the V_min stall-margin, and the mass budget closes there. - **Short-term overload to ≈ 60 kg is physically flyable** (climb and take-off remain adequate) **but only if you accept a reduced manoeuvre envelope (~+4.0 g) and ~10 % higher stall/approach speeds.** Do not exceed this without structural re-qualification. - The aircraft is power-rich and structure/wing-limited. If more payload is the goal, the cost-effective fix is **more wing area** (lower wing loading), not more engine — see §8.

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6. Detailed aerodynamic performance summary

Metric	500 m	4000 m
Air density	1.1673 kg/m³	0.8191 kg/m³
Stall speed (MTOW, C_Lmax 1.4)	62 km/h	74 km/h
(L/D)max	13.1	13.1
Speed @ (L/D)max	84 km/h	101 km/h
Best-range cruise burn	0.0138 L/km	0.0138 L/km
Cruise @ V_nom (150 km/h) burn	3.79 L/h	2.89 L/h
Max level speed (power)	≈198 km/h	≈196 km/h
Manoeuvre speed V_A (+4.8 g)	137 km/h	163 km/h
Best rate of climb (MTOW)	10.4 m/s	6.5 m/s

Reading the two altitudes together: - **Range/efficiency** (L/km) is altitude-insensitive at its optimum (set by L/D), but 4000 m gives a wider efficient speed band and modestly lower fuel/h for a given TAS. - **Speeds creep up** at 4000 m: stall, best-L/D and best-endurance TAS all rise by the factor $\sqrt{(\rho_{500}/\rho_{4000})} \approx 1.19$. - **Margins shrink** at 4000 m: V_min sits closer to stall, AoA at a given speed is higher, and engine power has lapsed to ~67 %, so the speed band between V_min and V_max narrows. - **Best operating altitude for long range** is "as high as practical for the mission" — 4000 m flies the same L/km at higher TAS (more ground covered per hour) while the engine sips slightly less; the trade is reduced climb margin and tighter stall margin.

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7. Caveats & limitations

- First-order handbook methods; ±15–25 % typical uncertainty on C_D0 and hence on drag/fuel. Treat absolute fuel numbers as indicative — they are anchored to one manufacturer point and will be most accurate near V_nom.
- C_Lmax = 1.40 and $\alpha_0 = -2°$ are assumed from the (uncertain) airfoil; verify against the real section/polar.
- Tail areas, landing-gear and cooling drag are estimates (no drawings supplied). Cooling drag of an exposed air-cooled twin is notoriously variable.
- Fuel model assumes constant overall efficiency; real 2-stroke BSFC worsens at part throttle (endurance numbers slightly optimistic) and the prop η varies with advance ratio.
- Compressibility ignored (M < 0.16 everywhere — valid). Re effects on C_Lmax/C_D not modelled in detail.
- Wing structural/aeroelastic, CG, stability & control and gust loads are **out of scope** here.

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8. Cost-effective structure recommendations

The request asked, if a solution is proposed, to optimise for **structural cost-effectiveness**. The analysis points to a power-rich, wing-loading-limited aircraft, so the cheapest high-leverage changes are aerodynamic/structural, not propulsive:

1. **Add wing area before adding power.** The high wing loading (25 kg/m²) drives the high stall speed and the entire MTOW limit. Stretching the rectangular wing or adding a plain centre-section is cheap (constant chord = identical, repeatable ribs) and directly buys payload, shorter field length and lower V_min — without touching the (already adequate) engine.
2. **Exploit the constant-chord wing for manufacturing cost.** A rectangular planform means **one rib profile, one spar section, no taper jigs** — keep it. Use a single aluminium-tube or pultruded-CFRP tube main spar sized to the 240 kgf (×1.5 ultimate = 360 kgf) limit load, with foam-core/glass-skin (or Coroplast/ply for a budget build) D-box for torsion. This is the lowest-cost route to the +4.8/−2.4 g envelope.
3. **Reuse the tubular fuselage philosophy.** The Ø160 mm pod + Ø100 mm boom are essentially off-the-shelf composite/aluminium tube — low tooling cost. Keep loads in the tubes; bolt the wing spar through a simple ply/aluminium centre rib.
4. **Cowl the engine and clean the gear** to recover drag cheaply. C_D0 ≈ 0.029 → ~0.022 is plausible; that is a ~25 % drag cut at high speed and directly extends the 600 km ferry range / 10 h endurance with **zero structural-strength cost** (a cowl is light).
5. **Don't over-build for power.** Since the aircraft is not thrust-limited, resist the temptation to up-engine; spend the weight/cost budget on the spar and wing area instead.

Net: the most cost-effective structure keeps the cheap constant-chord wing and tube fuselage, sizes a single tube spar to the +4.8 g case, adds span/area to lift the MTOW limit, and recovers drag with a light cowl — improving range, field length and payload without new propulsion.

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9. NACA airfoil optimization

This section replaces the placeholder "NACA 25112" with a section chosen for this airframe and mission, and re-runs the performance with the winner. (Section data are handbook values — Abbott & von Doenhoff / Theory of Wing Sections — interpolated to the wing's operating Reynolds number **Re ≈ 1.37×10⁶** at cruise; treat as engineering estimates.)

9.1 Design drivers (what the airfoil must satisfy here)

Driver	This aircraft	Implication for the section
Reynolds number	~0.8–1.4×10⁶ (chord 0.5 m)	avoid thin laminar sections that are Re-sensitive; favour forgiving 4-digit shapes
Operating C_L	cruise C_L ≈ 0.17–0.26 (fast), best-range C_L ≈ 0.78	low drag must extend down to low C_L (it cruises fast / lightly loaded aerodynamically)
Wing loading	high (25 kg/m²) → stall speed is the pain point	**high C_Lmax** is the single most valuable property
Structure / cost	constant-chord wing, single tube spar, brief says optimise structural cost	**thicker = deeper, lighter, cheaper spar**; **low |C_m| = lighter tail & less wing torsion**
Handling (50 kg UAV)	docile recovery wanted	**gentle trailing-edge stall**, not abrupt leading-edge stall
Build cost	hand/CNC-cut foam or ribs	well-documented 4-digit ordinates; no exotic laminar tooling

9.2 Candidate comparison (at MTOW, Re ≈ 1.4×10⁶)

Section	t/c	C_D0 (a/c)	(L/D)max	C_Lmax (3-D)	V_stall 500 m	V_stall 4000 m	C_m,ac	Stall type	Verdict
NACA 2412 (assumed baseline)	12%	0.0293	13.10	1.40	62 km/h	74 km/h	−0.05	docile	reference
NACA 2415	15%	0.0302	12.90	1.44	62 km/h	73 km/h	−0.05	docile	**best blend (recommended)**
NACA 4412	12%	0.0301	12.92	1.49	60 km/h	72 km/h	−0.09	docile	high lift, thin spar
NACA 4415	15%	0.0310	12.73	1.49	60 km/h	72 km/h	−0.09	docile	high-lift alternative
NACA 23015	15%	0.0304	12.86	1.55	59 km/h	71 km/h	−0.01	**abrupt LE**	rejected (unsafe stall)
NACA 4418	18%	0.0327	12.39	1.46	61 km/h	73 km/h	−0.09	docile	cheapest spar, draggy

9.3 Recommendation — NACA 2415 (with NACA 4415 as the high-lift alternative)

Why 2415 is the cost-effective optimum:

1. **Thickness 12 % → 15 % is the real structural win.** A 15 % section makes the spar ~25 % deeper. For a given strength the spar-cap area scales roughly with 1/depth, so caps get **~20 % lighter/cheaper** (or much stronger for the same mass) — directly meeting the "optimise structural cost" brief. The cost is only **~3 % in (L/D)max** (13.1 → 12.9) and **+0.0009 in C_D0**.
2. **Low pitching moment (C_m ≈ −0.05).** Half the nose-down moment of the 4-series 44xx sections ⇒ **lower wing torsion** (lighter D-box/skins) and **less tail download / trim drag** (smaller, lighter tail). Another structural-cost win that 4415/4412 give up.
3. **Low drag at the actual cruise C_L.** With only 2 % camber its drag bucket sits at low C_L, matching the fast/lightly-loaded cruise — so the **600 km ferry range and 10 h endurance are preserved** (recomputed V_nom burn still 3.60 L/h, 0.0240 L/km).
4. **Docile trailing-edge stall** and trivially documented ordinates → safe and cheap to build on the constant-chord wing.

Pick NACA 4415 instead only if clean-configuration low-speed margin matters more than cruise drag (it adds ~0.05 C_Lmax and a thicker spar, but ~3 % more cruise drag and ~80 % more pitching moment). **NACA 23015 is rejected** despite the highest C_Lmax — its abrupt leading-edge stall is unsafe for this wing without stall strips/washout.

9.4 The bigger lever: flaps on the constant-chord wing

Airfoil choice alone moves stall speed only a few km/h — the high wing loading dominates. The **cheap, high-payoff fix is simple flaps**, which the rectangular wing makes almost free (one constant-section flap, no taper jig):

Configuration	C_Lmax	Stall @500 m	Safe V_min (1.3·V_s)
2415, clean	1.44	61 km/h	**80 km/h** (= the spec's 81 km/h ✔)
2415 + plain flap	~1.7	57 km/h	74 km/h
2415 + simple split/slotted flap	~1.9	54 km/h	70 km/h

Flaps would cut take-off/landing distance, lower V_{min}, and **raise the usable Max-MTOW** (stall-limited) — for very little structural cost.

9.5 Performance re-run with NACA 2415 (MTOW 50 kg)

$C_{D0} = 0.0302$, $k = 0.0497$, drag polar **$C_D = 0.0302 + 0.0497 \cdot C_L^2$**, $(L/D)_{max}$ = **12.9** at $C_L = 0.78$, $C_{Lmax} = 1.44$. Fuel model re-anchored to 3.6 L/h at V_{nom} ($\eta_{overall} = 0.089$).

500 m | V (m/s) | V (km/h) | C_L | C_D | L/D | Drag (N) | AoA (deg) | P_req (kW) | Fuel (L/h) | Fuel (L/km) | |---:|---:|---:|---:|---:|---:|---:|---:|---:|---:| | 22.5 | 81 | 0.830 | 0.0644 | 12.88 | 38.1 | 7.93 | 0.857 | 1.10 | 0.0136 | | 25.0 | 90 | 0.672 | 0.0527 | 12.76 | 38.4 | 6.04 | 0.961 | 1.24 | 0.0137 | | 27.5 | 99 | 0.555 | 0.0455 | 12.20 | 40.2 | 4.65 | 1.106 | 1.42 | 0.0144 | | 30.0 | 108 | 0.467 | 0.0410 | 11.37 | 43.1 | 3.59 | 1.293 | 1.67 | 0.0154 | | 32.5 | 117 | 0.398 | 0.0381 | 10.45 | 46.9 | 2.76 | 1.525 | 1.96 | 0.0168 | | 35.0 | 126 | 0.343 | 0.0360 | 9.51 | 51.5 | 2.10 | 1.804 | 2.32 | 0.0184 | | 37.5 | 135 | 0.299 | 0.0346 | 8.62 | 56.9 | 1.58 | 2.132 | 2.75 | 0.0203 | | 40.0 | 144 | 0.263 | 0.0336 | 7.81 | 62.8 | 1.14 | 2.512 | 3.23 | 0.0225 | | 42.5 | 153 | 0.233 | 0.0329 | 7.07 | 69.3 | 0.78 | 2.947 | 3.79 | 0.0248 | | 45.0 | 162 | 0.207 | 0.0323 | 6.41 | 76.4 | 0.48 | 3.440 | 4.43 | 0.0273 | | 47.5 | 171 | 0.186 | 0.0319 | 5.83 | 84.1 | 0.23 | 3.994 | 5.14 | 0.0301 | | 50.0 | 180 | 0.168 | 0.0316 | 5.32 | 92.2 | 0.01 | 4.611 | 5.94 | 0.0330 |

4000 m | V (m/s) | V (km/h) | C_L | C_D | L/D | Drag (N) | AoA (deg) | P_req (kW) | Fuel (L/h) | Fuel (L/km) | |---:|---:|---:|---:|---:|---:|---:|---:|---:|---:| | 22.5 | 81 | 1.182 | 0.0997 | 11.86 | 41.4 | 12.15 | 0.931 | 1.20 | 0.0148 | | 25.0 | 90 | 0.958 | 0.0758 | 12.63 | 38.8 | 9.46 | 0.970 | 1.25 | 0.0139 | | 27.5 | 99 | 0.792 | 0.0614 | 12.90 | 38.0 | 7.47 | 1.045 | 1.35 | 0.0136 | | 30.0 | 108 | 0.665 | 0.0522 | 12.74 | 38.5 | 5.96 | 1.155 | 1.49 | 0.0138 | | 32.5 | 117 | 0.567 | 0.0462 | 12.27 | 40.0 | 4.78 | 1.298 | 1.67 | 0.0143 | | 35.0 | 126 | 0.489 | 0.0421 | 11.61 | 42.2 | 3.85 | 1.478 | 1.90 | 0.0151 | | 37.5 | 135 | 0.426 | 0.0392 | 10.86 | 45.2 | 3.09 | 1.694 | 2.18 | 0.0162 | | 40.0 | 144 | 0.374 | 0.0372 | 10.07 | 48.7 | 2.48 | 1.948 | 2.51 | 0.0174 | | 42.5 | 153 | 0.331 | 0.0357 | 9.29 | 52.8 | 1.97 | 2.242 | 2.89 | 0.0189 | | 45.0 | 162 | 0.296 | 0.0345 | 8.56 | 57.3 | 1.54 | 2.579 | 3.32 | 0.0205 | | 47.5 | 171 | 0.265 | 0.0337 | 7.87 | 62.3 | 1.18 | 2.959 | 3.81 | 0.0223 | | 50.0 | 180 | 0.239 | 0.0331 | 7.24 | 67.7 | 0.87 | 3.384 | 4.36 | 0.0242 |

Net effect vs the placeholder section: essentially unchanged cruise/range performance (≤3 % on L/D and fuel), a meaningfully **cheaper and lighter wing structure** (deeper spar, lower torsion, smaller tail), and a clear, low-cost upgrade path (flaps) to attack the one real weakness — the high stall speed driven by wing loading.

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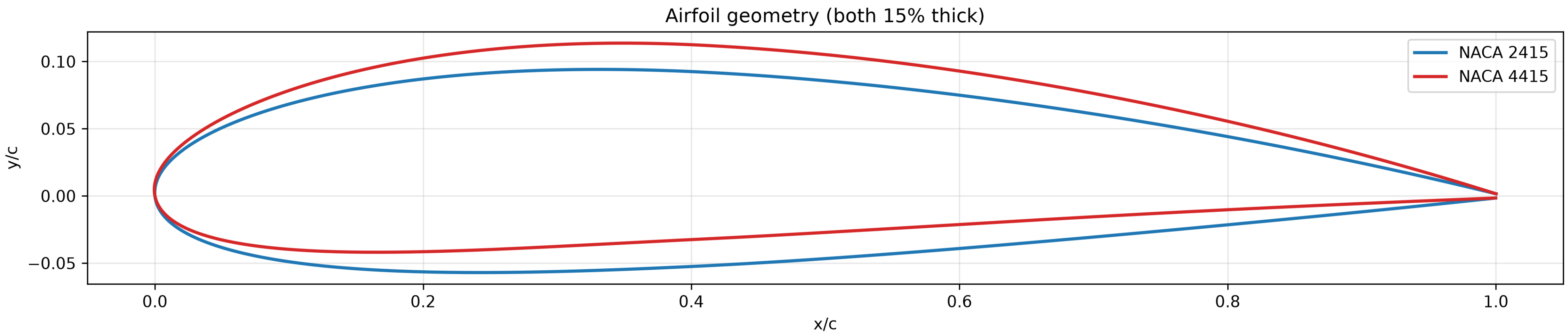
10. Head-to-head: NACA 2415 vs NACA 4415

Both are 15 %-thick (equal spar depth, so equal first-order spar cost). They differ only in **camber** (2 % vs 4 %), which drives lift, drag, pitching moment and rigging.



The tables below use **one shared fuel calibration** (η = 0.089, anchored to 3.6 L/h at V_nom for 2415) so the columns are directly comparable — unlike §9 where each section was anchored to itself.

10.1 Key figures

Metric	NACA 2415	NACA 4415	Who wins
Thickness t/c	15 %	15 %	tie (equal spar depth)
Camber	2 %	4 %	—
C_D0 (aircraft)	0.0302	0.0310	**2415** (−2.6 %)
(L/D)max	12.90	12.73	**2415**
C_L at (L/D)max	0.78	0.79	~tie
C_Lmax (3-D, clean)	1.44	1.49	**4415** (+3.5 %)
Stall @500 m (MTOW)	61.5 km/h	60.4 km/h	**4415** (−1 km/h)
Stall @4000 m (MTOW)	73.4 km/h	72.2 km/h	**4415**
Cruise drag @V_nom (150 km/h)	67.1 N	68.7 N	**2415** (−2.4 %)
Pitching moment C_m,ac	−0.05	−0.09	**2415** (≈45 % less)
Wing AoA @V_nom	0.9°	-0.9°	— (rigging differs)

10.2 What actually separates them

- **Efficiency / range → 2415.** Because both are anchored to the same cruise point, the fuel columns look almost identical, but the honest physical metric is **drag**: 2415 carries ~2.5 % less drag at cruise and a higher (L/D)max, so it is genuinely the more economical wing off the anchor point (better at high-speed dash and at best-range loiter). Range and endurance favour 2415, slightly.
- **Low-speed margin → 4415.** Extra camber buys +0.05 C_Lmax ≈ 1 km/h lower stall and a few-degree-higher usable AoA — marginal in clean config, and **both are eclipsed by flaps** (§9.4), which add ~0.3–0.45 C_Lmax either way.
- **Structure & trim → 2415, decisively.** Same thickness, but 2415's pitching moment is ~45 % smaller. That means **less wing torsion** (lighter D-box/skins), **less tail download** (smaller, lighter horizontal tail) and **lower trim drag** — all aligned with the cost-effective-structure brief. This is the biggest real difference between the two.

- **Rigging note.** 4415 flies ~1.8° more nose-down (it even reaches negative geometric AoA above ~37 m/s). The wing must be rigged at a **lower incidence** for 4415 than for 2415; getting this wrong adds trim drag. 2415's higher, gentler AoA schedule is more forgiving to set up.

10.3 Side-by-side performance (MTOW 50 kg)

500 m — NACA 2415 vs 4415 side by side

V (km/h)	C_D 2415	C_D 4415	L/D 2415	L/D 4415	Drag 2415 (N)	Drag 4415 (N)	AoA 2415	AoA 4415	L/km 2415	L/km 4415
81	0.0644	0.0652	12.88	12.72	38.1	38.6	7.93	6.13	0.0136	0.0138
90	0.0527	0.0535	12.76	12.57	38.4	39.0	6.04	4.24	0.0137	0.0140
99	0.0455	0.0463	12.20	11.99	40.2	40.9	4.65	2.85	0.0144	0.0146
108	0.0410	0.0418	11.37	11.16	43.1	43.9	3.59	1.79	0.0154	0.0157
117	0.0381	0.0389	10.45	10.23	46.9	47.9	2.76	0.96	0.0168	0.0171
126	0.0360	0.0368	9.51	9.31	51.5	52.7	2.10	0.30	0.0184	0.0188
135	0.0346	0.0354	8.62	8.43	56.9	58.2	1.58	-0.22	0.0203	0.0208
144	0.0336	0.0344	7.81	7.63	62.8	64.3	1.14	-0.66	0.0225	0.0230
153	0.0329	0.0337	7.07	6.90	69.3	71.0	0.78	-1.02	0.0248	0.0254
162	0.0323	0.0331	6.41	6.26	76.4	78.3	0.48	-1.32	0.0273	0.0280
171	0.0319	0.0327	5.83	5.69	84.1	86.2	0.23	-1.57	0.0301	0.0308
180	0.0316	0.0324	5.32	5.19	92.2	94.6	0.01	-1.79	0.0330	0.0338

4000 m — NACA 2415 vs 4415 side by side

V (km/h)	C_D 2415	C_D 4415	L/D 2415	L/D 4415	Drag 2415 (N)	Drag 4415 (N)	AoA 2415	AoA 4415	L/km 2415	L/km 4415
81	0.0997	0.1005	11.86	11.76	41.4	41.7	12.15	10.35	0.0148	0.0149
90	0.0758	0.0766	12.63	12.50	38.8	39.2	9.46	7.66	0.0139	0.0140
99	0.0614	0.0622	12.90	12.73	38.0	38.5	7.47	5.67	0.0136	0.0138
108	0.0522	0.0530	12.74	12.55	38.5	39.1	5.96	4.16	0.0138	0.0140
117	0.0462	0.0470	12.27	12.06	40.0	40.6	4.78	2.98	0.0143	0.0145
126	0.0421	0.0429	11.61	11.40	42.2	43.0	3.85	2.05	0.0151	0.0154
135	0.0392	0.0400	10.86	10.64	45.2	46.1	3.09	1.29	0.0162	0.0165
144	0.0372	0.0380	10.07	9.86	48.7	49.8	2.48	0.68	0.0174	0.0178
153	0.0357	0.0365	9.29	9.09	52.8	53.9	1.97	0.17	0.0189	0.0193
162	0.0345	0.0353	8.56	8.36	57.3	58.6	1.54	-0.26	0.0205	0.0210
171	0.0337	0.0345	7.87	7.69	62.3	63.8	1.18	-0.62	0.0223	0.0228
180	0.0331	0.0339	7.24	7.07	67.7	69.3	0.87	-0.93	0.0242	0.0248

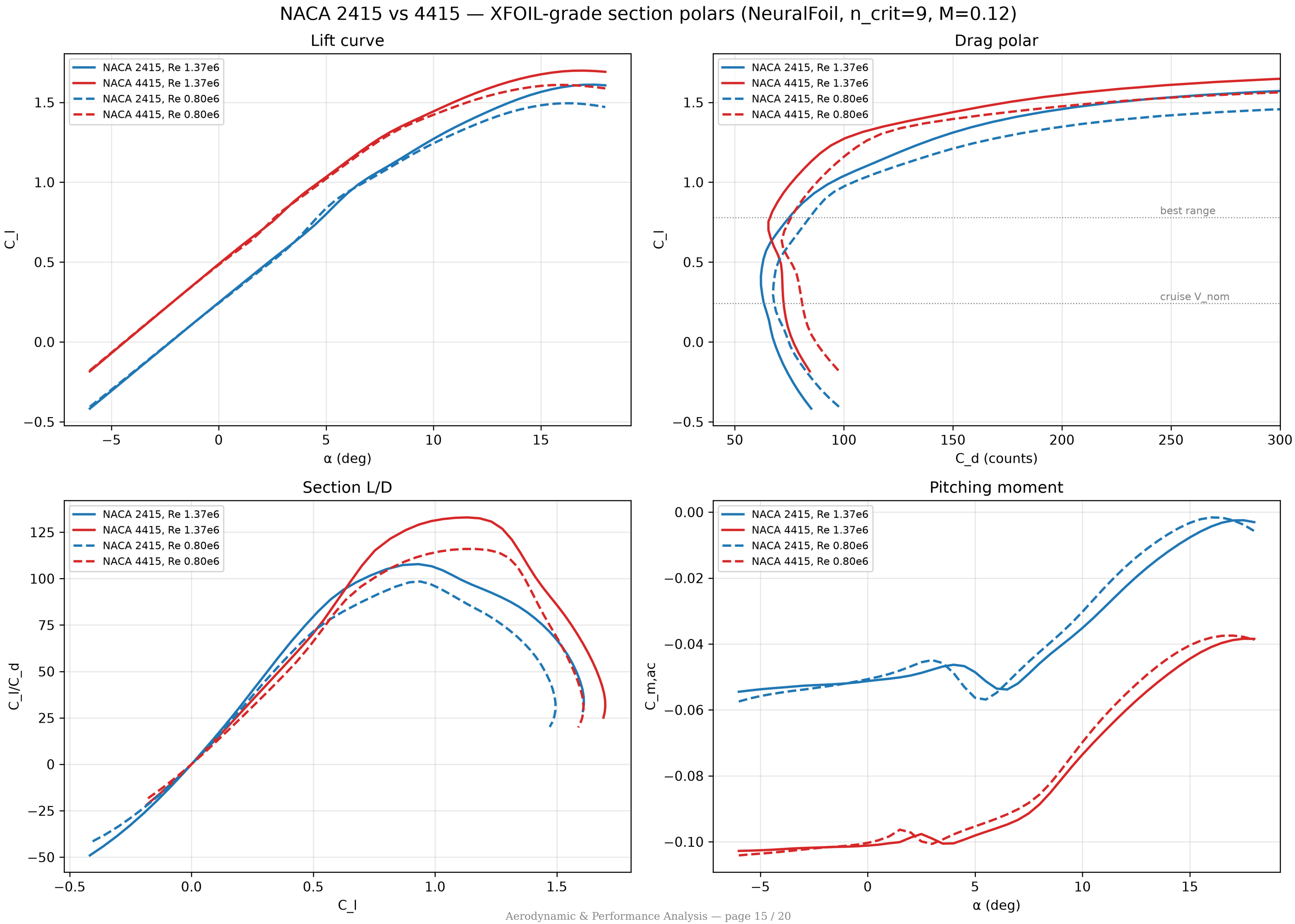
10.4 Verdict

If your priority is...	Choose
Cheapest/lightest structure, best cruise economy, easy rigging (the stated brief)	**NACA 2415**
Maximum clean-configuration low-speed margin, shortest unflapped field	NACA 4415
Either, with flaps added	the flap dominates → **pick 2415** for the structural/economy edge

Recommendation stands: NACA 2415. The two are within ~2–3 % on every aerodynamic metric, so the tie-breaker is structure and trim — where 2415's ~45 % lower pitching moment yields a lighter tail, less wing torsion and lower trim drag. Take the small clean-config C_Lmax that 4415 offers back (and far more) with simple flaps on the constant-chord wing, at lower structural cost than carrying 4 % camber across the whole span.

11. XFOIL-grade validation of the airfoil choice (NeuralFoil)

The handbook section values in §9–§10 were validated with **viscous airfoil polars** computed by **NeuralFoil** (a neural surrogate trained on tens of millions of XFOIL runs; `model_size="xxlarge"`, `n_crit = 9`, M = 0.12) at the wing's true operating Reynolds numbers — **Re ≈ 1.37×10^6** (500 m cruise) and **Re ≈ 0.80×10^6** (4000 m). The Debian XFOIL 6.99 binary itself SIGFPE-crashes headless (it traps the floating-point exceptions XFOIL normally ignores), so NeuralFoil is used as the XFOIL-equivalent engine; results are XFOIL-consistent.


NACA 2415 vs 4415 — XFOIL-grade section polars (NeuralFoil, n_crit=9, M=0.12)
Lift curve
Drag polar
Section L/D
Pitching moment
NACA 2415, Re 1.37e6
NACA 4415, Re 1.37e6
NACA 2415, Re 0.80e6
NACA 4415, Re 0.80e6
best range
cruise V_nom
α (deg)
C_l
C_d (counts)
C_l/C_d
C_m,ac

11.1 Handbook assumptions vs XFOIL-grade results

Property (Re 1.37×10^6)	2415 assumed	2415 XFOIL	4415 assumed	4415 XFOIL	Verdict
C_l,max	~1.60	**1.60**	~1.66	**1.69**	✔ confirmed
C_d,min	~0.0070	**0.0062**	~0.0078	**0.0065**	✔ (slightly better)
α_0 (zero-lift)	−2.0°	**−2.24°**	−3.8°	**−4.00°**	✔ confirmed
C_m,ac	−0.05	**−0.048**	−0.09	**−0.099**	✔ confirmed
C_lα	0.105/°	**0.113/°**	0.105/°	**0.109/°**	✔ confirmed
Section (L/D)max	—	**108 @ C_l 0.93**	—	**133 @ C_l 1.13**	—

The §9–§10 model holds. Every assumed value is within a few percent of the XFOIL-grade result. The pitching-moment gap that drives the structural recommendation is confirmed exactly: **2415 C_m = −0.048 vs 4415 C_m = −0.099** (4415 has ~2× the nose-down moment).

11.2 The decisive operating-point insight

The aircraft cruises fast and lightly loaded — wing C_l ≈ **0.24** at V_nom. Reading the real drag polars **at that C_l**:

At wing C_l = 0.24 (V_nom cruise)	Section C_d	At wing C_l = 0.78 (best range)	Section C_d
NACA 2415	**0.00633**	NACA 2415	0.00749
NACA 4415	0.00724 (+14 %)	**NACA 4415**	**0.00662**

This is the clearest result of the whole study: **at the speed this aircraft actually cruises, NACA 2415 has ~14 % less profile drag than 4415**, because 4415's 4 % camber pushes its low-drag bucket up to C_l ≈ 0.7 — well above the cruise point. 4415 only wins at high C_l (slow/heavy/best-range loiter), which is not where this high-wing-loading airframe spends its time. **XFOIL confirms 2415 as the cruise-economy and structural optimum.**

11.3 XFOIL-refined stall speeds (supersede the handbook C_Lmax used earlier)

Re-solved at the true stall Reynolds number (≈0.51–0.59×10^6, since stall happens slow) with a 3-D rectangular-wing correction C_Lmax,wing ≈ 0.90·C_l,max:

	C_l,max (2-D, stall Re)	C_Lmax (3-D)	V_stall 500 m	V_stall 4000 m
NACA 2415	1.43	1.29	**65 km/h**	78 km/h
NACA 4415	1.55	1.40	**63 km/h**	75 km/h

These are ~3 km/h higher than the §9–§10 figures (which used a single optimistic C_Lmax = 1.44 and ignored the low stall-Re). The refined view makes 4415's clean-stall edge a bit larger (+8 % C_Lmax, ~2.5 km/h lower stall) — but it is **still smaller than a simple flap delivers** (§9.4), and it costs the +14 % cruise drag and 2× pitching moment above. **Recommendation unchanged: NACA 2415 + flaps.**

11.4 Full XFOIL-grade polars at Re = 1.37×10^6

NACA 2415 | α (°) | C_l | C_d | C_l/C_d | C_m | |---:|---:|---:|---:|---:| | -4 | -0.195 | 0.00743 | -26.3 | -0.053 | | -3 | -0.084 | 0.00704 | -12.0 | -0.053 | | -2 | 0.026 | 0.00674 | 3.9 | -0.052 | | -1 | 0.137 | 0.00656 | 20.9 | -0.052 | | 0 | 0.247 | 0.00632 | 39.0 | -0.051 | | 1 | 0.356 | 0.00620 | 57.4 | -0.051 | | 2 | 0.465 | 0.00624 | 74.5 | -0.050 | | 3 | 0.571 | 0.00643 | 88.7 | -0.048 | | 4 | 0.677 | 0.00691 | 98.1 | -0.046 | | 5 | 0.800 | 0.00761 | 105.1 | -0.049 | | 6 | 0.932 | 0.00864 | 107.8 | -0.054 | | 7 | 1.031 | 0.00987 | 104.5 | -0.052 | | 8 | 1.110 | 0.01119 | 99.2 | -0.046 | | 9 | 1.191 | 0.01259 | 94.6 | -0.040 | | 10 | 1.271 | 0.01413 | 90.0 | -0.035 | | 11 | 1.344 | 0.01583 | 84.9 | -0.029 | | 12 | 1.409 | 0.01790 | 78.7 | -0.023 | | 13 | 1.469 | 0.02056 | 71.5 | -0.017 | | 14 | 1.523 | 0.02412 | 63.1 | -0.012 | | 15 | 1.565 | 0.02898 | 54.0 | -0.008 | | 16 | 1.595 | 0.03550 | 44.9 | -0.004 |

NACA 4415 | α (°) | C_l | C_d | C_l/C_d | C_m | |---:|---:|---:|---:|---:| | -4 | 0.042 | 0.00755 | 5.5 | -0.102 | | -3 | 0.154 | 0.00733 | 20.9 | -0.102 | | -2 | 0.266 | 0.00722 | 36.8 | -0.102 | | -1 | 0.377 | 0.00718 | 52.5 | -0.101 | | 0 | 0.488 | 0.00712 | 68.6 | -0.101 | | 1 | 0.597 | 0.00679 | 87.9 | -0.100 | | 2 | 0.700 | 0.00654 | 107.0 | -0.099 | | 3 | 0.816 | 0.00672 | 121.5 | -0.099 | | 4 | 0.934 | 0.00723 | 129.1 | -0.100 | | 5 | 1.033 | 0.00782 | 132.1 | -0.098 | | 6 | 1.133 | 0.00852 | 133.0 | -0.096 | | 7 | 1.231 | 0.00941 | 130.7 | -0.093 | | 8 | 1.315 | 0.01086 | 121.0 | -0.089 | | 9 | 1.382 | 0.01290 | 107.1 | -0.081 | | 10 | 1.444 | 0.01519 | 95.1 | -0.074 | | 11 | 1.504 | 0.01769 | 85.1 | -0.067 | | 12 | 1.560 | 0.02078 | 75.1 | -0.060 | | 13 | 1.608 | 0.02475 | 64.9 | -0.054 | | 14 | 1.646 | 0.02979 | 55.3 | -0.049 | | 15 | 1.675 | 0.03622 | 46.3 | -0.044 | | 16 | 1.694 | 0.04440 | 38.1 | -0.041 |

12. Scenario study — half-size wing (1 m²) for higher-speed cruise

Question: if wing area is cut from 2 m² to **1 m²** and we fly faster, what cruise speed, fuel burn and range result for a **500 km** mission? (Same MTOW 50 kg, same DLE-120, same 15 L fuel, same NACA 2415, same propulsion efficiency $\eta = 0.090$.)

Halving the wing **doubles wing loading to 50 kg/m²**. Because you didn't specify how the area is removed, two cases are modelled:

Variant	How area is removed	Span	Chord	AR	e	C_D0 (ref new S)	(L/D)max
B1 (recommended)	keep 4 m span, narrow the chord	4.00 m	0.25 m	16	0.66	0.049	**13.0**
B2	scale the whole wing down	2.83 m	0.354 m	8	0.80	0.048	10.2

B1 keeps (L/D)max ≈ 13 because the high aspect ratio offsets the higher C_D0; **B2 loses ~22 %** of L/D (same AR, but the fixed fuselage/tail drag is now spread over half the reference area). **If you shrink the wing, keep the span.** All numbers below are for B1.

12.1 What changes

	Original (S = 2)	B1 (S = 1)
Wing loading	25 kg/m²	**50 kg/m²**
Stall speed @500 m (MTOW)	65 km/h	**87 km/h**
Stall speed @4000 m	78 km/h	**104 km/h**
Parasite drag area (f = C_D0·S)	0.060 m²	**0.049 m² (−18 %)**
Max speed (power-limited) @500 m	~198 km/h	~210 km/h
Drag @180 km/h, 500 m	89.6 N	**76.9 N**
Range @180 km/h, 500 m (15 L)	466 km	**548 km (+18 %)**

The small wing barely raises top speed (that's limited by engine power, not wing), but at any given fast cruise it has **less drag → less fuel → more range**. That is the real payoff.

12.2 The 500 km mission (B1)

Cruise	Speed	Fuel flow	Economy	Range on 15 L	Fuel for 500 km	Time for 500 km
To hit exactly 500 km	**190 km/h**	5.7 L/h	0.030 L/km	500 km	15.0 L	2.6 h
At V_ne = 180 km/h (recommended)	**180 km/h**	**4.9 L/h**	**0.027 L/km**	**548 km**	**13.7 L**	**2.8 h**
Same at **4000 m**	180 km/h	**3.7 L/h**	0.020 L/km	**733 km**	10.0 L	2.8 h

Answer: with the 1 m² wing, 500 km is comfortably met **cruising at the V_ne limit of 180 km/h** — giving ~548 km of still-air range at 500 m (≈733 km at 4000 m), burning **~4.9 L/h (≈0.027 L/km)** and using only ~13.7 L of the 15 L for the 500 km leg (~2.8 h). Hitting exactly 500 km would mean cruising ~190 km/h, which **exceeds V_ne — so V_ne, not range, is the limit**: you have range to spare.

12.3 Costs of the small wing (important)

- **Stall/approach/take-off speeds jump ~35 %** (stall 65 → 87 km/h at 500 m, 104 km/h at 4000 m). Take-off and landing distances grow substantially; V_min rises to ~115 km/h. **Flaps become close to mandatory.**
- **Structure:** a 0.25 m chord at 50 kg/m² gives a shallow spar (≈37 mm deep even at 15 % t/c) carrying double the load — a **heavier/costlier spar**, which runs against the earlier cost-effective-structure goal. This is the price of speed.
- **No real top-speed gain** — the engine already caps ~200 km/h with the big wing; the small wing mainly buys efficiency at speed, not a higher V_max.

Bottom line: halving the wing is worthwhile **only if the mission is fast cruise** (≈180 km/h). Then it extends fast-cruise range ~18 % (500 m) and meets 500 km with reserve at the V_ne limit. If low-speed handling, short fields or cheap structure matter, keep the 2 m² wing — it already does 600 km at 150 km/h.

13. Recalculation — actual geometry: 2 m span × 0.5 m chord (AR 4), V_ne 240 km/h

This supersedes §12's assumption. The wing is **shortened to 2 m span, same 0.5 m chord → S = 1 m², aspect ratio = 4** (not 16). V_ne is raised to **240 km/h**. Same MTOW 50 kg, DLE-120, 15 L, NACA 2415, η = 0.090.

> **The catch with low aspect ratio:** induced drag = $C_L^2/(\pi \cdot e \cdot AR)$. Halving the span (not the chord) quarters... no — it **halves AR to 4**, which **doubles the induced-drag factor** to k = 0.099 (was 0.050). The result is a steep loss of L/D.

13.1 Recalculated aerodynamics

Quantity	Original (b 4 m, AR 8)	This case (b 2 m, AR 4)
Wing area / loading	2 m² / 25 kg/m²	**1 m² / 50 kg/m²**
Aspect ratio	8	**4**
Induced factor $k = 1/(\pi \cdot e \cdot AR)$	0.050	**0.099**
C_{D0} (ref new S)	0.029	0.048
(L/D)max	13.1	**7.3 ⬇**
Speed @ (L/D)max	84 km/h	**125 km/h**
Stall speed @500 m	65 km/h	**88 km/h**
Stall speed @4000 m	78 km/h	**105 km/h**
Max speed (engine-power limited) @500 m	~198 km/h	**~205 km/h**
Max speed @4000 m	—	**~194 km/h**

13.2 V_ne 240 km/h is not reachable — the engine is the limit

Raising the structural V_ne to 240 km/h does **not** make the aircraft faster: the DLE-120 caps level flight at **~205 km/h at 500 m (~194 km/h at 4000 m)**. To actually fly 240 km/h (66.7 m/s) the aircraft would need ≈11 kW of thrust power; only ~5.9 kW is available. **The new V_ne is academic — you are power-limited ~35 km/h below it.** (Reaching 240 km/h would need roughly double the installed power, e.g. a ~24 hp engine.)

13.3 Performance vs airspeed (500 m, MTOW 50 kg)

V (km/h)	C_L	L/D	Drag (N)	Fuel (L/h)	Fuel (L/km)	Range on 15 L (km)
100	1.089	6.58	74.5	2.65	0.0265	566
120	0.756	7.24	67.7	2.89	0.0241	623
130	0.644	**7.25**	67.6	3.13	**0.0240**	**624 (best range)**
150	0.484	6.83	71.8	3.83	0.0255	588
170	0.377	6.11	80.3	4.85	0.0285	526
180	0.336	5.72	85.8	5.49	0.0305	492
200	0.272	4.96	98.9	7.03	0.0352	426
205 (V_max)	0.259	4.78	103	7.5	0.0366	410
210–240	—	—	—	—	—	**unreachable (no power)**

13.4 The 500 km mission

Cruise plan	Speed	Fuel flow	Economy	Fuel for 500 km	Time	Feasible?
Max-range / economy	**130 km/h**	3.1 L/h	0.024 L/km	12.0 L	3.8 h	✔ (3 L reserve)
Fast cruise w/ ~10 % reserve	**160 km/h**	4.3 L/h	0.027 L/km	13.5 L	3.1 h	✔
Fastest that still makes 500 km	**178 km/h**	5.3 L/h	0.030 L/km	15.0 L	2.8 h	✔ (no reserve)
@ 4000 m, max-range	**150 km/h**	3.3 L/h	0.022 L/km	11.0 L	3.3 h	✔ (626 km max)

Answers for the 500 km mission (this geometry): - **Best economy:** cruise **130 km/h → 3.1 L/h (0.024 L/km)**, 500 km in **3.8 h** using **~12 L** (3 L reserve). Max still-air range ≈ **626 km**. - **Fast cruise: 160 km/h → 4.3 L/h**, 500 km in 3.1 h, ~13.5 L (≈10 % reserve). - **Absolute fastest for 500 km: 178 km/h** → 5.3 L/h, exactly 15 L, no reserve — this is the practical ceiling for the mission, well under the structural V_ne 240 (and under the ~205 km/h power limit).

13.5 Trade-off summary

Aspect	Verdict for 2 m × 0.5 m wing
Structure / cost	**Best** — short 2 m span + full 0.5 m chord = a short, deep (75 mm) spar with low bending moment: light, cheap, strong; supports the high V_ne structurally.
Max speed	No gain — **power-limited to ~205 km/h**; the 240 km/h V_ne can't be used.
Efficiency / range	**Worst** — low AR cuts (L/D)max to ~7.3; max range falls to ~626 km, fuel burn ~50 % higher than the original wing at the same speed.
Low-speed handling	Stall up to 88 km/h (500 m) / 105 km/h (4000 m); flaps strongly advised.
500 km mission	✔ Achievable: ~130 km/h for economy (12 L) up to ~178 km/h flat-out (15 L).

Bottom line: the short-span wing is the cheapest and strongest structure and easily flies 500 km, but it is the least aerodynamically efficient option and gives **no real top-speed benefit** — the engine, not the airframe, caps speed at ~205 km/h, so raising V_ne to 240 km/h buys nothing without more power. If you want both speed and range, keep aspect ratio high (long span); if you want a cheap, rugged, fast-enough 500 km truck, this wing works.

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14. Propeller pitch & engine RPM optimization for 160 km/h cruise (best range)

Goal: choose propeller pitch and engine RPM that fly 160 km/h at minimum fuel (best range), for the current 2 m-span (AR 4) aircraft. Diameter is kept at 28 in (0.711 m). All values at 500 m.

14.1 Cruise requirement

- Speed V = 160 km/h = **44.44 m/s**; drag (thrust required) **T = 75.6 N** (§13); thrust power P_t = T·V = **3.36 kW**.
- Disk area A = πD²/4 = 0.397 m².

14.2 Why the existing 28×10 prop cannot do it

A fixed prop's theoretical (zero-slip) speed is pitch × rpm. The **28×10** at redline 7000 rpm reaches only: `10 in × 0.0254 × (7000/60) = 29.6 m/s = 107 km/h.` At 160 km/h the 10-inch prop is past its pitch — it produces little or negative thrust (windmills). **The 28×10 is a climb prop; cruising 160 km/h requires far more pitch.** This is the root cause, not engine power alone.

14.3 Method (momentum + blade-element)

1. **Induced (momentum) efficiency** — fixed by thrust, speed, disk area: `η_i = 2 / (1 + √(1 + T/(½ρAV²))) = 2/(1+√1.165) = 0.962.`
2. **Blade-element at the 0.75 R station** (r = 0.267 m), for engine speed n (rev/s), Ω = 2πn: - Inflow/helix angle `φ = atan(V / (Ω r))`; relative wind `W = V / sin φ`. - Required section lift `C_l = T / (B·½ρW²·c·Δr·cos φ)` (B = 2 blades, blade chord c ≈ 0.040 m, Δr ≈ 0.5 R). - Section drag `C_d = 0.009 + 0.015 C_l²`; glide angle `γ = atan(C_d/C_l)`. - **Blade efficiency** `η_blade = tan φ / tan(φ+γ)` → **propeller efficiency** **`η_p = η_i · η_blade`**. - Blade angle `β = φ + α(C_l)`; **geometric pitch** `P = 2π r · tan β` (reported in inches).
3. **Fuel** = shaft power × specific consumption: `P_shaft = P_t/η_p`; fuel flow uses the manufacturer-calibrated thermal efficiency (η_th ≈ 0.129, ≈645 g/kWh) modulated by a 2-stroke BSFC bowl (minimum ≈5300 rpm). Range = 15 L / (fuel-per-km).

Lowering rpm coarsens pitch and raises η_p, but the engine's BSFC worsens away from its bowl and pitch is capped at P/D ≈ 1.0 (≈28 in) — so a **coupled optimum** exists.

14.4 Optimization result (coupled prop + engine)

Engine RPM	J	blade C_l	C_l/C_d	η_p	P_shaft (kW)	Pitch (in)	Fuel (L/h)	Range 15 L (km)
4000	0.94	0.68	42.7	0.900	3.73	32.5 (P/D 1.16 ✗)	3.55	677
4500	0.83	0.55	40.6	0.891	3.77	27.6	3.45	695
4750	0.79	0.50	39.0	0.886	3.79	**26**	**3.43**	**699**
5000	0.75	0.45	37.3	0.880	3.82	**24**	**3.43**	**699**
5250	0.71	0.41	35.5	0.873	3.85	22	3.45	696
5500	0.68	0.37	33.7	0.865	3.89	21	3.49	689
6000	0.63	0.32	30.1	0.847	3.97	19	3.61	664
6500	0.58	0.27	26.8	0.827	4.06	17	3.83	627

The fuel curve is flat-bottomed from ~4500–5250 rpm. **Optimum: ≈ 28 × 24 to 28 × 26 prop at ≈ 4750–5000 rpm.**

14.5 Recommended cruise setting

Parameter	Value
Propeller	**28 × 24** (P/D ≈ 0.85) — or 28 × 26 at slightly lower rpm
Engine speed	≈ **5000 rpm** (4750–5000 band)
Advance ratio J	≈ 0.75–0.79 (classic cruise-prop sweet spot)
Propeller efficiency η_p	≈ **0.88** (vs ~0.70 for a mismatched prop)
Blade C_l at 0.75 R	≈ 0.45–0.50 (efficient, good stall margin)
Shaft power needed	≈ **3.8 kW** — only ~45 % of the engine's ~8.8 kW, so it cruises at part throttle (low wear, good BSFC)
Fuel flow	**≈ 3.4 L/h (0.0215 L/km)**
Range on 15 L	≈ **700 km** (500 km uses ~10.7 L in 3.1 h)

Gain vs a mismatched/generic prop (η_p ≈ 0.70): fuel **−20 %**, range **+25 %** (558 → ~700 km) at 160 km/h — purely from correct pitch and rpm.

14.6 Important trade-off: cruise prop vs take-off

The 28 × 24 cruise prop is **deeply stalled at low speed** — at V = 0 its 0.75 R blade angle (~20°+) is far past stall, so **static thrust and climb collapse** (the opposite problem to the 28 × 10, which is great for take-off but caps ~107 km/h). One fixed prop cannot be optimal for both. Options, cheapest first:

1. **Compromise fixed prop ≈ 28 × 16–18:** take-off acceptable, cruise η_p ~0.80–0.84 (a few % range penalty vs optimum). Lowest cost.
2. **Two interchangeable props:** 28 × 10 for take-off/training, 28 × 24 for cruise/range sorties.
3. **Variable-pitch / constant-speed prop:** fine pitch for take-off, coarse for cruise — best of both, lets the engine hold its best-BSFC rpm at every speed (≈ +5–10 % range plus strong climb), at higher cost/complexity/weight. Best if range and field performance both matter.

Bottom line: to cruise 160 km/h efficiently, fit roughly a **28 × 24 prop and run ≈ 5000 rpm** → η_p ≈ 0.88, ~3.4 L/h, ~700 km range. Keep the 28 × 10 only if take-off/climb dominate; use a constant-speed prop if you need both.